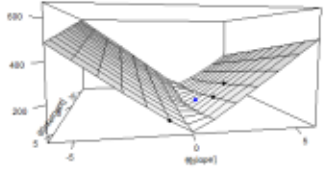
Introduction to Machine Learning

ML-Basics
Losses & Risk Minimization



Learning goals

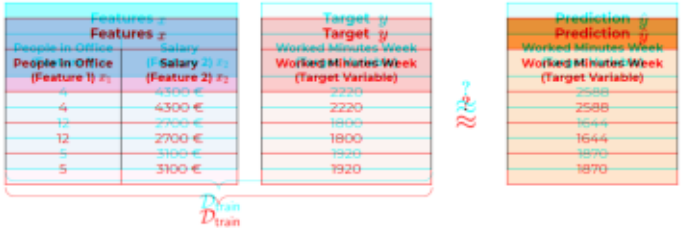
- Know the concept of loss
- Understand the relationship between loss and risk
- Understand the relationship between risk minimization and finding the best model

Learning goals

- Know concept of loss function
- Understand relationship between loss and risk
- Understand relationship between risk minimization and finding best model

HOW TO EVALUATE MODELS

- When training a learner, we optimize over the hypothesis space, to find the function that fits the training data best.
- We are looking for a function, with which the predicted output per training point is as close as possible to the observed label.



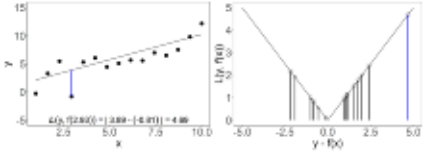
- To make this precise, we need to define now how we measure the difference between a prediction and a ground truth label pointwise.
- We need to measure this pointwise difference precisely.

LOSS

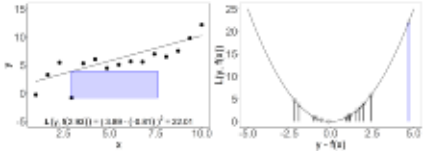
The **loss function** $L(y, f(\mathbf{x}))$ quantifies the "quality" of the prediction $f(\mathbf{x})$ of a single observation $\mathbf{x}$:

$$L : \mathcal{Y} \times \mathbb{R}^g \to \mathbb{R}.$$

In regression, we can use the absolute loss $L(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$:



or the L2-loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$:



RISK OF A MODEL

- The (theoretical) **risk** associated with a certain hypothesis $f(\mathbf{x})$ measured by a loss function $L(y, f(\mathbf{x}))$ is the **expected loss**

$$\mathcal{R}(f) := \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] = \int L(y, f(\mathbf{x}))\, \mathrm{d}\mathbb{P}_{xy}.$$

- This is the average error we incur when we use f on data from $\mathbb{P}_{xy}$.
- Goal in ML: Find a hypothesis $f(\mathbf{x}) \in \mathcal{H}$ that **minimizes** the risk.

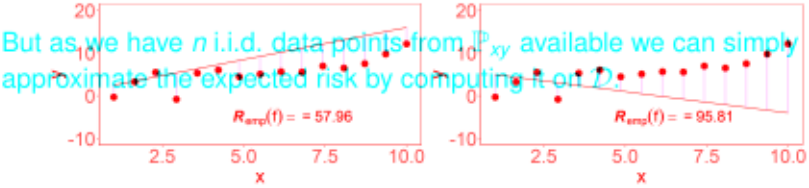
EMPIRICAL RISK

To evaluate, how well a given function f matches our training data, we now simply sum up all pointwise losses.

$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

This **empirical risk function** allows us to associate one quality score with each $f \in \mathcal{H}$, which encodes how well it fits the training data.

$$\mathcal{R}_{\text{emp}} : \mathcal{H} \to \mathbb{R}$$



EMPIRICAL RISK MINIMIZATION

To evaluate, how well a given function f matches our training data, we now simply sum-up all f's pointwise losses.

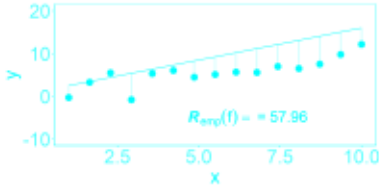
- The best model is the model with the smallest risk.
- For finite $\mathcal{H}$, we could tabulate all models and select the best.

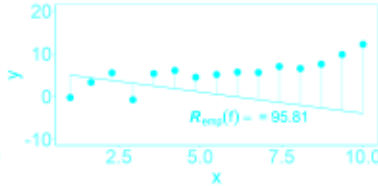
$$\mathcal{R}_{\text{emp}}(f) = \sum^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\theta)$
f_1	2	3	194.62
f_2	3	2	127.12
f_3	6	-1	95.81
f_4	1	1.5	57.96

$$\mathcal{R}_{\text{emp}} : \mathcal{H} \to \mathbb{R}$$

This gives rise to the **empirical risk function** which allows us to associate one quality score with each of our models, which encodes how well our model fits our training data.





EMPIRICAL RISK MINIMIZATION

- But usually $\mathcal{H}$ is infinitely large.
- Instead we can consider the risk surface w.r.t. the parameters $\boldsymbol{\theta}$ (i.e., the visualization of $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$).

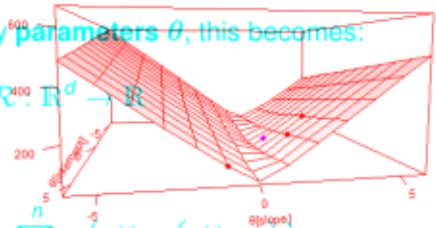
$$\mathcal{R}_{\text{emp}}(f) = \frac{1}{n}\sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right).$$

The factor $\frac{1}{n}$ does not make a difference in optimization, so we will consider $\mathcal{R}_{\text{emp}}(f)$ most of the time.

$$\mathcal{R}_{\text{emp}} : \mathbb{R}^d \to \mathbb{R}$$

- Since f is usually defined by parameters $\boldsymbol{\theta}$, this becomes:

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$
f_1	2	3	194.62
f_2	3	2	127.12
f_3	6	-1	95.81
f_4	1	1.5	57.96



$$\mathcal{R} : \mathbb{R}^d \to \mathbb{R}$$

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

EMPIRICAL RISK MINIMIZATION

The best model is the model with the smallest risk.

If we have a finite number of models f, we could simply tabulate them and select the best.

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\theta)$
f_1	2	3	194.62
f_2	3	2	127.12
f_3	6	-1	95.81
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EMPIRICAL RISK MINIMIZATION

But usually $\mathcal{H}$ is infinitely large.

Instead we can consider the risk surface w.r.t. the parameters θ.
(By this I simply mean the visualization of $\mathcal{R}_{\text{emp}}(\theta)$)

$$\mathcal{R}_{\text{emp}}(\theta) : \mathbb{R}^d \to \mathbb{R}.$$

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\theta)$
f_1	2	3	194.62
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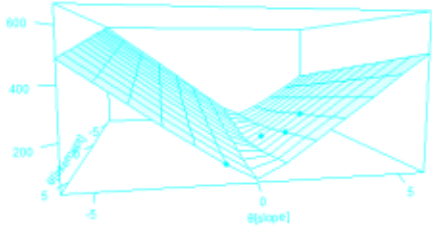


EMPIRICAL RISK MINIMIZATION

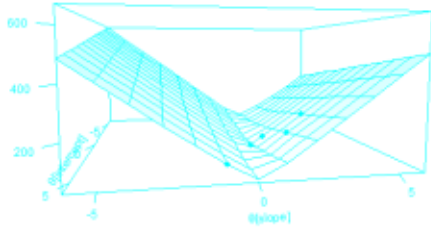
Minimizing this surface is called **empirical risk minimization** (ERM).

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta).$$

Usually we do this by numerical optimization.

$$\mathcal{R} : \mathbb{R}^d \to \mathbb{R}.$$

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\theta)$
f_1	2	3	194.62
f_2	3	2	127.12
f_3	6	-1	95.81
f_4	1	1.5	57.96
f_5	1.25	0.90	23.40



In a certain sense, we have now reduced the problem of learning to **numerical parameter optimization**.